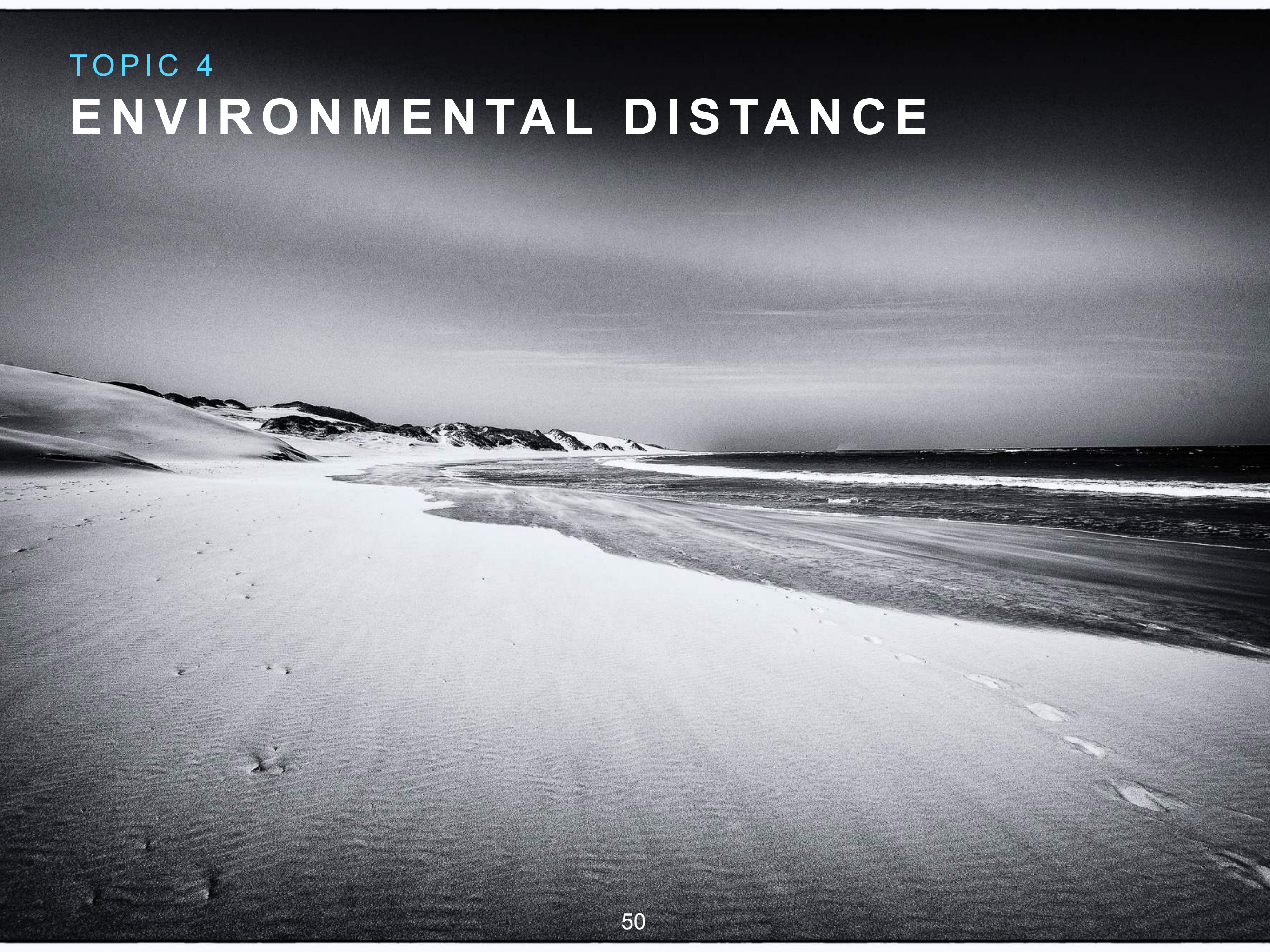


TOPIC 4

ENVIRONMENTAL DISTANCE



Similarity/dissimilarity and distance

- Species:
 - sites sharing a similar species composition are ecologically similar
 - i.e. high similarity / low dissimilarity;
 - we will cover dissimilarities in Topic 5.
- Environmental variables:
 - sites sharing similar environmental conditions have a low environmental distance between them;
 - how similar sites to each other depends on...
 - measurable environmental differences that influence species composition,
 - it can be due to unmeasured influences, or
 - it can also simply be noise.
- It is the ecologist's role to figure out what influences the similarity/dissimilarity/distances between sites.

Distance matrices

- A **distance matrix** is produced from a data table (**species table** or **environment table**) by calculating one of several dissimilarity indices.
- Also called **association** or **resemblance** matrices.
- See `vegdist()` **vegan** for a list of dissimilarity indices.
- The result is a matrix of **pairwise differences** in community composition (as synthesised by the chosen index) or ecological distance between all sites.

Distance matrix for environmental data

- **Euclidian distance** is “the ‘ordinary’ straight-line distance between two points in Euclidean space” (i.e. in its simplest form a planar area, which you know of as a graph with x- and y-axes)
- So, in 2D and 3D, gives distance in **cartesian units** between points on a plane (x, y) or in a volume (x, y, z).
- Conforms to our physical concept of distance
 - e.g. short geographic distances between points on a map, and
 - (loses accuracy over large distances, as Earth’s surface is not on a plane but on a sphere... correct by using great circle distances, e.g. use the Haversine formula).
- Calculated using the Pythagorean theorem
 - differences are squared, so single large differences become very important, and
 - this is not useful for species data.

Distance matrices

- The matrices are **square** and **symmetrical**, and they will have as many rows and columns as the number of sites present in the original species or environment table.
- The **diagonals are zero** (a site is the same as itself, so it has zero dissimilarity).
- The table can be read directly, and each cell represents the species or ecological difference between a pair of sites.
- All information of the species ID (and maybe also abundance) of a site is lost, as this info is condensed into one metric.

Two dimensions [[edit](#)]

In the [Euclidean plane](#), if $\mathbf{p} = (p_1, p_2)$ and $\mathbf{q} = (q_1, q_2)$ then the distance is given by

$$d(\mathbf{p}, \mathbf{q}) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2}.$$

This is equivalent to the [Pythagorean theorem](#).

Alternatively, it follows from (2) that if the [polar coordinates](#) of the point $\mathbf{p}$ are (r_1, θ_1) and those of $\mathbf{q}$ are (r_2, θ_2) , then the distance between the points is

$$\sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_1 - \theta_2)}.$$

Three dimensions [[edit](#)]

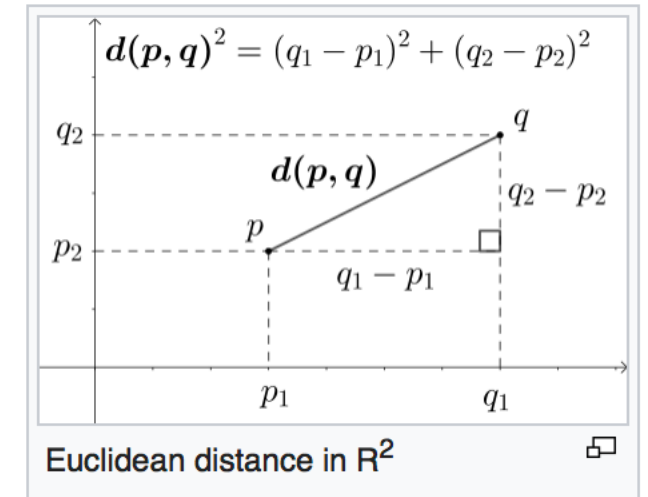
In three-dimensional Euclidean space, the distance is

$$d(\mathbf{p}, \mathbf{q}) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + (p_3 - q_3)^2}.$$

n dimensions [[edit](#)]

In general, for an n -dimensional space, the distance is

$$d(\mathbf{p}, \mathbf{q}) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + \cdots + (p_i - q_i)^2 + \cdots + (p_n - q_n)^2}.$$



e.g. example with position (such as geographic) coordinates...
use **vegan**'s `vegdist()` function

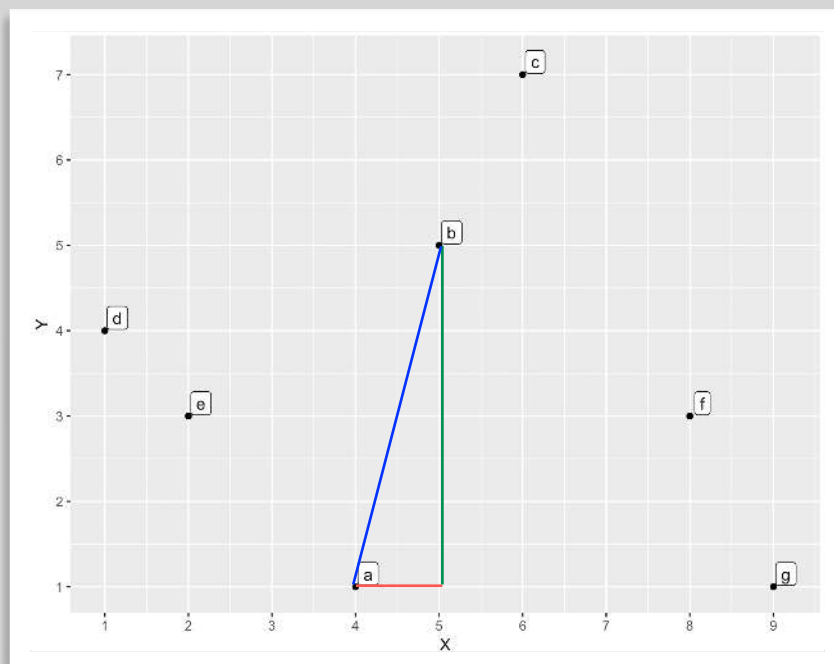
Raw data

site	x	y
a	4	1
b	5	5
c	6	6
d	1	4
e	2	3
f	8	3
g	9	1

Euclidian distances

```
> ex.xy.euc ← vegdist(ex.xy, method = "euclidian")
> ex.xy.euc
```

	a	b	c	d	e	f
b	4.123106					
c	6.324555	2.236068				
d	4.242641	4.123106	5.830952			
e	2.828427	3.605551	5.656854	1.414214		
f	4.472136	3.605551	4.472136	7.071068	6.000000	
g	5.000000	5.656854	6.708204	8.544004	7.280110	2.236068



$$d(a, b) = \sqrt{(a_x - b_x)^2 + (a_y - b_y)^2}$$

e.g. example with 3D position coordinates (a.k.a. dimensions)...

Raw data

site	x	y	z
a	4	1	3
b	5	5	5
c	6	6	4
d	1	4	9
e	2	3	8
f	8	3	1
g	9	1	5

Euclidian distances

```
R> ex.xyz.euc <- vegdist(ex.xyz[,2:4], method = "euclidian")
R> ex.xyz.euc
```

	1	2	3	4	5	6
2	4.582576					
3	5.477226	1.732051				
4	7.348469	5.744563	7.348469			
5	5.744563	4.690416	6.403124	1.732051		
6	4.898979	5.385165	4.690416	10.677078	9.219544	
7	5.385165	5.656854	5.916080	9.433981	7.874008	4.582576

$$d(a, b) = \sqrt{(a_x - b_x)^2 + (a_y - b_y)^2 + (a_z - b_z)^2}$$

e.g. example with environmental 'dimensions'...

Raw data

site	temperature	depth	light
a	4	1	3
b	5	5	5
c	6	6	4
d	1	4	9
e	2	3	8
f	8	3	1
g	9	1	5

Euclidian distances

```
R> ex.xyz.euc <- vegdist(ex.xyz[,2:4], method = "euclidian")
R> ex.xyz.euc
```

	1	2	3	4	5	6
2	4.582576					
3	5.477226	1.732051				
4	7.348469	5.744563	7.348469			
5	5.744563	4.690416	6.403124	1.732051		
6	4.898979	5.385165	4.690416	10.677078	9.219544	
7	5.385165	5.656854	5.916080	9.433981	7.874008	4.582576

$$d(a, b) = \sqrt{(a_{\text{temp}} - b_{\text{temp}})^2 + (a_{\text{depth}} - b_{\text{depth}})^2 + (a_{\text{light}} - b_{\text{light}})^2}$$

e.g. example with higher dimension environmental data...

Raw data

	pH	O2	temp	depth
a	7.1	6.5	12.1	1.1
b	7.5	5.5	12.3	1.3
c	7.6	5.7	11.9	1.5
d	7.0	5.4	11.8	1.6
e	7.1	6.3	12.0	1.8
f	7.2	6.3	12.1	1.9
g	6.9	6.1	12.2	2.2

(transformation)

Standardised data

```
> ex.env.std <- decostand(xy.env, method = "standardize")
> ex.env.std
```

	pH	O2	temp	depth
a	-0.3872983	1.2156767	0.2494233	-1.41749621
b	1.1618950	-1.0842522	1.4133987	-0.88114629
c	1.5491933	-0.6242664	-0.9145521	-0.34479637
d	-0.7745967	-1.3142450	-1.4965398	-0.07662142
e	-0.3872983	0.7556909	-0.3325644	0.45972850
f	0.0000000	0.7556909	0.2494233	0.72790346
g	-1.1618950	0.2957051	0.8314110	1.53242833

Euclidian distances

```
> ex.env.euc <- vegdist(xy.datex.env.std, method = "euclidian")
> ex.env.euc
```

	a	b	c	d	e	f
b	4.123106					
c	6.324555	2.236068				
d	4.242641	4.123106	5.830952			
e	2.828427	3.605551	5.656854	1.414214		
f	4.472136	3.605551	4.472136	7.071068	6.000000	
g	5.000000	5.656854	6.708204	8.544004	7.280110	2.236068